

Monalisa Dokania

Software Engineer

Location: Tempe, Arizona, USA · Open to relocate

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GitHub: github.com/msdokania

Portfolio: msdokania.github.io

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PROFESSIONAL PROFILE

Software engineer focused on **AI agents, LLM systems, distributed systems, and cloud infrastructure**. Production experience across backend systems, event-driven architectures, and AWS/EKS, alongside hands-on LLM work in RAG, agentic workflows, evaluation pipelines, and AI copilots. Founding-engineer experience shipping products from architecture to launch.

TECHNICAL SKILLS

Languages	TypeScript · Python · Go · Java · JavaScript
Backend	Node.js · tRPC · FastAPI · Spring Boot · Go Microservices · PostgreSQL · Redis · Kafka · SQS
Frontend	Next.js · React · TypeScript · Vite · Tailwind · HTML/CSS · Vercel · SSE
AI / ML	RAG · Agentic AI · LLM Applications/Vertical Agents · Tool use/MCP · LangChain · LangGraph · Evals · Vector DBs · Streaming Chat UIs · Prompt Engineering · Machine Learning models
Cloud & Infra	AWS · EKS · Kubernetes · Docker · Terraform · CI/CD · Prometheus · Grafana
Tools	Git · GitHub Actions · JIRA · Linux · Testing frameworks · AI dev tools (Cursor, Copilot, etc.)

WORK EXPERIENCE

06/2024 – present

Scottsdale, AZ, USA

Lead Software Engineer

BLUSVN Technologies

- **Founding engineer** on a distributed water-delivery platform; owned the **full product lifecycle** from architecture to production across **backend, infrastructure, and ML**.
- Architected and shipped **100+ backend APIs** (TypeScript/tRPC, PostgreSQL) powering customer, driver, and admin applications; reduced end-to-end latency by **40%**.
- Redesigned the **Next.js admin panel UI** for business admins and managers; Built new operational dashboards and management features that streamlined daily workflows.
- Engineered high-performance **Golang microservices** with concurrent worker pools and batched notification delivery for order allocation and driver dispatch workflows.
- Built real-time **ML inference pipelines** for face/ID validation and **AI-assisted agent workflows** for log analysis and reporting that cut manual investigation effort during incident triage.
- Built **Kafka event-driven** location workflow (Kafka-Redis pipeline) for real-time driver tracking.
- Designed and operated production **AWS/EKS infrastructure** with Terraform; reduced AWS spend by **12%** and led **load and chaos testing** to validate SLOs under peak load.
- Partnered with product managers and stakeholders on technical architecture and **led engineering delivery; managed CI/CD and Agile sprints**.

05/2023 – 05/2024

Tempe, AZ, USA

Graduate Teaching Assistant

Arizona State University

- Supported Software Verification, Validation & Testing — **900+ students** across sections.
- Guided students in unit testing, debugging, code reviews, and OOP design principles.
- Led lab sessions and office hours; developed course materials and grading rubrics.

07/2016 – 05/2022

India

Business Operations Analyst

M/S Rajiv Industries - Family-owned manufacturing business

- Supported day-to-day operations and reporting workflows to track sales, inventory, customer orders, and SKU-level performance at a family-run manufacturing business. Converted historical records into internal datasets for data retrieval, demand analysis and stock-planning.
- Returned to full-time technical work through an M.S. in Computer Science in 2022.

06/2015 – 06/2016

Noida, India

Associate Software Engineer

Indus Valley Partners Pvt. Ltd.

- Developed **custom applications on the Geneva financial platform**, a portfolio management system used by asset managers; Performed statistical reconciliation across portfolios.
- Contributed to a major platform upgrade (Geneva 8.0 → 15.2), including data migration validation.

TECHNICAL PROJECTS

04/2026 – present

Live Suggestions Copilot: Real-time AI Meeting Assistant

- **Tech Stack:** *Next.js, TypeScript, Groq, Whisper, MediaRecorder API, SSE, Tailwind*

Real-time AI copilot that transcribes live meeting audio and generates context-aware suggestions, with a streaming chat panel for follow-up answers. Engineered prompt rules for topic locking, recency weighting, anti-repetition, suggestion diversity, and hallucination control, plus error handling for robust streaming.

 <https://github.com/msdokania/live-suggestions-copilot>

 **Demo:** <https://live-suggestions-copilot.vercel.app/>

12/2025 – present

Research Assistant: Hybrid RAG + RAGAS Evaluation

- **Tech Stack:** *Python, LangChain, ChromaDB, BM25, Cross-encoders, RAGAS, OpenAI, FastAPI*

Production-grade research assistant built in four parts: naive RAG baseline → advanced retrieval (hybrid semantic + BM25, cross-encoder re-ranking, citation enforcement) → RAGAS-based evaluation against 60 ground-truth Q&A pairs measuring faithfulness, answer relevancy, and context precision/recall → systematic prompt iteration with measured improvements.

Part of my Agentic AI Series published on LinkedIn and GitHub.

 <https://github.com/msdokania/agentic-ai-series/tree/master/research-assistant-rag>

10/2025 – present

Multi-Agent LLM Pipeline for Document & Log Analysis

- **Tech Stack:** *Python, LangChain, FastAPI, Qdrant DB, Redis RQ, OpenAI*

Five-stage multi-agent system (Planner → Retriever → Analyst → Critic → Reporter) for document and log analysis. Full RAG workflow with semantic retrieval, async PDF processing via Redis RQ, confidence scoring, and a feedback loop where the Critic agent catches ~30% of unsupported claims before they reach the final report — measurably fewer hallucinations than single-prompt approaches.

 <https://github.com/msdokania/agentic-analysis-platform>

EDUCATION

08/2022 – 05/2024

Tempe, AZ, USA

Master of Science, Computer Science

Arizona State University

Final grade: 4.0 / 4.0

2011 – 2015

Allahabad, India

Bachelor of Technology, Information Technology

Motilal Nehru National Institute of Technology (MNNIT Allahabad)